MIL-F-3617D

8 October 1975
SUPERSEDING
MIL-F-3617(WC)
4 October 1971

MILITARY SPECIFICATION

# FIRING MECHANISM, ROCKET: ELECTRICAL

(FOR 3.5 INCH ROCKET LAUNCHER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of trigger-operated electrical firing mechanism for 3.5 inch rocket launchers.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | | |
|---|---|---|
| VV-L-800 | - | Lubricating Oil, General Purpose, Preservative (Water Displacing, Low Temperature). |

Military

| | | |
|---|---|---|
| MIL-G-10944 | - | Gages, Dimensional Control |
| MIL-P-14232 | - | Parts, Equipment and Tools for Army Materiel, Packaging and Packing of. |
| MIL-I-45607 | - | Inspection Equipment, Acquisition, Maintenance and Disposition of. |

STANDARDS

Military

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-109 | - | Quality Assurance Terms and Definitions. |
| MIL-STD-130 | - | Identification Marking of U.S. Military Property. |

FSC 1055

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DRAWINGS

Rock Island Arsenal
C7140325 - Mechanism, Electrical Firing, Group Assembly.

(Product drawings referenced in this specification form a part of C7140325).

PUBLICATIONS

United States Government
SF32 - General Provisions

Rock Island Arsenal
P7140325 - Packaging Data Sheet for Mechanism, Electrical Firing, Group Assembly.
DL7140325 - Quality Engineering Data List.

(Acceptance inspection equipment drawings referenced in this specification form a part of DL7140325).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The firing mechanism shall conform to the material, construction and design requirements specified herein, and on Drawing C7140325.

3.2.1 Coil assembly. The coil wire shall be wound continuously without joints or splices. The insulating sleeves, at the coil ends, shall be secure within the coil bobbin. The leads shall be brought out of the bobbin in a manner to insure against breakage due to strain or vibration.

3.2.1.1 Armature. The armature shall function without snag or bind within the coil bobbin.

3.2.2 Handgrips. When assembled, there shall be no looseness between the mechanism and the handgrips. The opening between the mating surfaces of the two grips shall not exceed 0.026 inches.



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3.3 Performance characteristics.

3.3.1 Safety button. The safety button shall have a positive holding action in both the SAFE and FIRE positions, and shall slide without binding to and from each of these positions. In the SAFE position, the button shall prevent any actuation of the mechanism. With the button in the FIRE position the trigger shall have free movement, through its complete range of travel. Testing shall be as specified in 4.5.3.2.

3.3.2 Trigger pull. The trigger shall operate with a smooth continuous movement. The trigger pull necessary to operate the firing mechanism shall be not less than 7 pounds and not more than 14 pounds, at the tip of the trigger. Testing shall be as specified in 4.5.3.3.

3.3.3 Screw adjustment. With the firing mechanism attached to a 3.5 inch Rocket Launcher or suitable test fixture, adjusting the screw, Drawing A7141213, shall obtain maximum output. There shall only be one perceptible click when the trigger is activated. Minimum acceptable output shall be 48 milliwatt seconds. Testing shall be as specified in 4.5.3.2.

3.3.4 Reliability. The electrical firing mechanism shall demonstrate a Mean-Actuation-To-Failure in excess of 500 actuations. Testing shall be as specified in 4.4.

* 3.3.5 Assembleability. The firing mechanism shall readily assemble to 3.5 inch Rocket Launchers.

3.4 Cleaning and lubrication. All parts of the firing mechanism shall be clean and entirely free of foreign matter, which may have accumulated due to magnetic attraction. Moving parts with metal to metal contact shall be free of fingerprints or other corrosive residues. After cleaning, metal parts shall be preserved by wiping with lubricating oil conforming to VV-L-800. All pivot points shall be lubricated with the same type oil.

3.5 Marking. The firing mechanism shall be marked as specified on the applicable drawing and in accordance with MIL-STD-130.

* 3.6 Workmanship. Workmanship shall be in accordance with the highest grade practice used in manufacturing military weapons. Finished items and parts shall not exhibit poor material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations and missing operations which may affect serviceability, functioning, operation or safety. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the reliability test specified in 4.6.3, and other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lots. Unless otherwise specified (see 6.1), the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105.

4.5.2 Examination, inspection and test provisions. Quality assurance provisions used in conjunction with this specification specify those product characteristics considered to be minimum inspection requirements for acceptability to assure conformance with performance and interchangeability requirements of the item.

4.5.2.1 In-process inspection. In-process inspection shall be established at strategically located points throughout the manufacturing processes to assure continuous control of product quality. With the exception of unmodified standard and commercial parts, the inspection system shall provide for inspection and approval of the first piece at each operation, or the finished part, before quantity production. The contractor shall provide and maintain work gages and other measuring and testing devices necessary to accomplish inspection and control quality during the manufacturing process.



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4.5.2.1.1 The Quality Assurance provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspection of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. In any event, the Government reserves all rights under General Provisions, Standard Form 32 (Inspection Article 5).

4.5.2.2 Inspection records. Records of inspection shall be accurate, complete and available upon request. The records of inspection shall provide, as a minimum, the following information:

a. Contract order number.
b. Drawing and specification number.
c. Nomenclature.
d. Number of pieces inspected.
e. Individual characteristics inspected.
f. Pass/fail criteria.
g. Description and quantity of defects found.
h. Results of inspection. Actual inspection and test results shall be recorded when actual measurements and tests were basis for pass/fail decisions.
i. Date of inspection.
j. The following shall be included when sampling is used:

(1) Sampling plan.
(2) Acceptable Quality Levels (AQLs) used.
(3) Lot size and lot number.
(4) Sample size.

4.5.2.2.1 Records of qualifications of operators for special processes (i.e., magnetic particle, radiographic, etc.), shall be made available. This shall apply no matter where the special processes are performed. All inspection records, including certified test reports, operator/process certifications, test reports, reports of visits to subcontractors, etc., shall be maintained on file and be made readily available upon request, for the duration of the contract and for three (3) years thereafter.

4.5.2.3 Certification provisions.

4.5.2.3.1 Certified test reports (CTR). A certified test report shall be made available for each lot of parts, assemblies, sub-systems and systems by lot number prior to acceptance. CTRs are not required for Military Standards (MS) and commercial items. This test report is in addition to and not in lieu of any rights of the Government under this



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contract or law. A CTR may be used as an element incident to, but shall not be used as the sole basis for, Government acceptance of the contract item(s). As a minimum, the report shall contain the following:

a. Name of company and date.
b. Contract number or purchase order number, national stock number and drawing number.
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment will be given.
d. All inspection and tests required by contract shall be recorded in test reports. These reports shall identify each lot, submitted for acceptance by lot number, the specification or drawing, revision and date, grade or type as applicable, number of specimens tested, specified characteristics and requirements, and actual results obtained.
e. Reports of the raw material producer's chemical and physical analysis.
f. A statement, as follows, certifying that material meets all requirements of the contract:

'The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: ALL of the information supplied above is true and accurate; the material covered by this certificate conforms to all contract requirements (including but not limited to the drawings and specifications); the inspection and test results, and the analysis appearing herein are true and accurate; and this certificate is made for the purpose of inducing payment and with the knowledge that the information and certification may be used as a basis for payment.
g. Signature and title of certifying official.

4.5.2.3.2 Certificate of conformance (COC). A certificate of conformance is required for Military Standard (MS) and commercial items. When a CTR is not required for material and process specification requirements, a COC, supported by inspection and test data, material analysis, or certification from the raw material producer or processor shall be made available for the specification covering raw material, processed material and processes. The COC shall be made available to the Government prior to or with the request to perform acceptance inspection equipment. This is in addition to and not in lieu of any rights of the Government. A COC may be used as an element incident to, but shall not be used as the sole basis for Government acceptance of contract item(s). As a minimum, the COC shall contain the following:

a. Name of company and date.



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b. Contract number or purchase order number, national stock number and drawing number.
c. Complete nomenclature of supplies together with lot number or other identification. The quantity in each lot or shipment will be given.
d. A statement, as follows, certifying that material meets all requirements of the contract:

'The undersigned individually, and as the authorized representative of the contractor, warrants and represents that: ALL of the information supplied above is true and accurate; the material covered by the certificate conforms to all contract requirements; the analysis appearing herein is a true and accurate analysis; and this certificate is made for the purpose of inducing payment and with the knowledge that the information and certification may be used as a basis for such payment.

e. Signature and title of certifying official.

4.5.2.4 Classification of defects. The following provisions shall be applicable to the prescribed inspection methods. Requests for a method other than that specified shall be submitted for Government approval (see 4.5.2.4.4 below):

4.5.2.4.1 Where "Visual" is specified as the inspection method for protective coating, the coating shall be visually examined for completeness, uniformity in appearance and color, and for freedom from pits, corrosion, scratches, and worn or bare spots.

4.5.2.4.2 Where "Visual" is specified as the inspection method for assemblies, the assembly shall be visually examined for completeness, security, function and conformance to specified requirement.

4.5.2.4.3 Where "SMTE" (Standard Measuring and Test Equipment) is specified as the method of inspection, the contractor may use any type of industry-developed, commerically available, multi-usage equipment or special inspection equipment approved by the Government.

4.5.2.4.4 Alternative inspection methods and inspection equipment may be used by the contractor when such methods and equipment equal or exceed the specified accuracy and provide, as a minimum, the quality assurance required in the contractual documents. Prior to applying such alternative inspection methods and inspection equipment, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the Procuring Contracting Officer (PCO) that their effectiveness is equal to or better than the contractual quality assurance method or equipment.



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4.5.2.4.5 Electrical firing mechanism group assembly (Drawing C7140325.

Examination Provisions

| Class | Characteristics | AQL | Inspection Method |
|---|---|---|---|
| 1 | Nonconformance of coil assembly (see 3.2.1). | 1.5 | Visual |
| 2 | Assembleability with launcher (see 3.3.5). | 1.5 | D8431365 |
| 3 | Presence of magnetically attracted foreign matter (see 3.4). | 1.5 | Visual |
| 4 | Nonconformance of handgrip to 3.2.2. | 1.5 | SMTE |
| 5 | Identification marking missing or incorrect (see 3.5). | 4.0 | Visual |
| 6 | Workmanship (see 3.6). | 4.0 | Visual |

4.5.2.5 Parts. Examination of parts shall be as specified in the contract (see 6.1), examination or these parts shall be prior to assembly into the end item or acceptance as repair parts. Only those parts found to be acceptable shall be used in assemblies.

4.5.3 Testing.

* 4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete firing mechanism. If test requirements cited herein are not met, acceptance of the mechanism shall be deferred and the supplier shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine firing mechanisms, partially assembled mechanisms, and components (including components and subassemblies at in-process of final assembly) to assure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot.

* 4.5.3.2 Safety button and screw adjustment testing. A random sample of firing mechanisms shall be selected from each inspection lot in accordance with MIL-STD-105. The firing mechanism shall be tested for safety



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button (see 3.3.2) and screw adjustment (see 3.3.4) using the test method specified in 4.6.1. Acceptance of the represented lot shall be based on an AQL of 0.65 percent defective.

* 4.5.3.3 Trigger pull testing. A random sample of firing mechanisms shall be selected from each inspection lot in accordance with MIL-STD-105. The firing mechanism shall be tested for the requirements of 3.3.2 using the test method specified in 4.6.2. Acceptance of the represented lot shall be based on an AQL of 0.65 percent defective.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.1), the packaging and marking examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquisition, calibration, maintenance and disposition. Unless otherwise specified (see 6.1), responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on the Quality Engineering Data List DL7140325 and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607 and MIL-G-10944.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of ten percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

* 4.6.1 Safety button and screw adjustment test. The firing mechanism shall be tested using the Electrical Output Tester, Drawing A8431328. The tester shall be connected to the test assembly, Drawing B8432716, and the frame (ground). Test the firing mechanism for a total of three (3) cycles as follows:

a. Place the safety button in FIRE position.
b. Slowly squeeze the trigger and ascertain that the output of the firing mechanism is at least 48 milliwatt seconds.
c. As the trigger is squeezed, ascertain that only one click is perceptible.
d. Place the safety button in SAFE position. Slowly squeeze the trigger to ascertain that the trigger is firmly blocked and resists squeezing action, also that there is no actuation of the firing mechanism or movement of the tester pointer.

4.6.2 Trigger pull test. The firing mechanism shall be mounted in a stationary position. Using a spring scale or other suitable device, approved by the Government representative, capable of registering in excess of 14 pounds. The trigger pull is to be measured at the tip of the trigger.



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4.6.3 Reliability test. The firing mechanism shall be suitably mounted so that the trigger can be mechanically activated at a rate of not less than five nor more than ten cycles per minute. To demonstrate the required Mean-Actuation-To-Failure (see 3.3.4), the trigger shall be actuated 1,500 times, and tested in accordance with 4.6.1 and 4.6.2 after each 300 actuations. Readjustment of the firing mechanism for "single click" is permissible.

4.6.4 Assembleability test. The firing mechanism shall be tested for assembleability by securing it to a test fixture conforming to Drawing D8431365.

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. The pilot pack shall consist of a complete firing mechanism packaged in accordance with Packaging Data Sheet P7140325 for the level of protection specified in the contract (see 6.1), packed level B and forwarded in accordance with 3.1.

* 5.2 Preservation, packaging, packing and marking. The firing mechanism shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P7140325 for the level of protection specified in the contract (see 6.1).

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the firing mechanism, showing applicable revision dates.
c. Requirements for submission of first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Inspection lot requirements, if different (see 4.5.1).
e. Examination criteria for components (see 4.5.2.5).
f. List of inspection equipment to be furnished the contractor (see 4.5.5) and responsibilities for other Government property to be furnished the contractor.
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing shall be specified on a DD Form 1423 included in the contract.

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, or the Management Control Systems Summary List, DD Form 1660.



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6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - WC

User activity:
Navy - MC

Preparing activity:
Army - WC

Project number:
1055-A054



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-F-3617D; Firing Mechanism, Rocket: Electrical (For 3.5 Inch Rocker Launcher)

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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